

Aircraft Structural Repair: Standards for Metallurgy, Composite Architecture, and Airworthiness Compliance

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Chapter 1

Structural Repair Architecture, Regulatory Framework, and ATA Alignment

The structural airworthiness of transport category aircraft is governed by the stringent requirements of 14 CFR Part 25, specifically Subpart C (Structure), which mandates that the airframe must withstand limit loads without detrimental permanent deformation and ultimate loads (defined as 1.5 times the limit loads) without structural failure. The maintenance, inspection, and repair of these primary and secondary load-bearing structures are categorized under the Air Transport Association (ATA) iSpec 2200 standard, encompassing ATA Chapter 51 (Standard Practices and Structures General), Chapter 52 (Doors), Chapter 53 (Fuselage), Chapter 54 (Nacelles/Pylons), Chapter 55 (Stabilizers), Chapter 56 (Windows), and Chapter 57 (Wings). The foundational philosophy governing modern structural design is damage tolerance, codified in 14 CFR 25.571, which assumes that initial flaws or manufacturing defects exist within the structure and mandates that the airframe must sustain these flaws, as they grow due to cyclic fatigue, corrosion, or accidental damage, without catastrophic failure until the flaw is detected during scheduled inspections. This paradigm supersedes the legacy safe-life and fail-safe design philosophies, requiring maintenance engineering to utilize fracture mechanics to calculate the critical crack length a_c and the subsequent inspection intervals. The fatigue crack growth rate is mathematically modeled using the Paris-Erdogan law, expressed as $\frac{da}{dN} = C(\Delta K)^m$, where a is the crack length, N is the number of load cycles, C and m are material-specific empirical constants, and ΔK is the stress intensity factor range, defined as $\Delta K = Y\Delta\sigma\sqrt{\pi a}$, with Y representing the geometric correction factor and $\Delta\sigma$ the applied cyclic stress range.

Structural repairs must restore the original static strength, fatigue life, and damage tolerance characteristics of the parent structure. The Federal Aviation Administration (FAA) Advisory Circular AC 43.13-1B (Acceptable Methods, Techniques, and Practices - Aircraft Inspection and Repair) and AC 43.13-2B (Acceptable Alterations) provide the baseline approved data for minor repairs. However, major structural repairs, defined as those that might appreciably affect weight, balance, structural strength, performance, or flight characteristics, require specific approval via the aircraft manufacturer's Structural Repair Manual (SRM) or a Designated Engineering Representative (DER) approved Form 8110-3. The transfer of aerodynamic and inertial loads through a repaired joint is highly dependent on the geometric continuity and the stiffness

matching of the repair doubler to the parent skin. A mismatch in axial stiffness (EA) or bending stiffness (EI) induces severe stress concentrations at the edges of the repair patch, accelerating fatigue crack initiation. The stress concentration factor K_t at the edge of a circular cutout or repair boundary is theoretically 3.0 for an isotropic infinite plate under uniaxial tension, but this value is heavily modified by the proximity of adjacent structural members, the fastener hole pattern, and the anisotropic nature of composite laminates. Consequently, structural repair engineers must perform rigorous finite element analysis (FEA) and classical hand calculations to verify that the bearing, bypass, and net-tension stresses within the repair zone remain below the allowable limits defined in MIL-HDBK-5 (Metallic Materials and Elements for Aerospace Vehicle Structures) or CMH-17 (Composite Materials Handbook).

Chapter 2

Metallurgical Properties and Heat Treatment of Aerospace Aluminum Alloys

Aerospace aluminum alloys are predominantly selected from the 2000-series (aluminum-copper) and 7000-series (aluminum-zinc) families due to their exceptional specific strength and fracture toughness. The 2024 alloy, typically utilized in the T3 or T351 temper, exhibits superior fatigue crack propagation resistance and is the standard material for fuselage skin and lower wing tension caps where cyclic tensile loads dominate. Conversely, the 7075 alloy in the T6 temper provides maximum yield strength but is highly susceptible to stress corrosion cracking (SCC) and exfoliation corrosion in the short transverse (ST) grain direction. To mitigate SCC vulnerabilities in thick sections and highly stressed components, the 7000-series alloys are overaged to the T73, T7351, or T76 tempers, which precipitates the strengthening phases (such as η' and η) more coarsely and discontinuously along the grain boundaries, thereby disrupting the continuous anodic path required for intergranular corrosion propagation. The mechanical properties of these alloys are highly anisotropic; the longitudinal (L) direction exhibits the highest tensile strength, the long transverse (LT) direction exhibits intermediate strength, and the short transverse (ST) direction exhibits the lowest strength and highest susceptibility to delamination. Structural repair designs must strictly account for this anisotropy, ensuring that primary tensile loads are aligned with the L or LT grain directions of the repair doubler.

To protect the underlying high-strength alloy from galvanic and pitting corrosion, aluminum sheets are frequently manufactured with a metallurgically bonded surface layer of high-purity aluminum, known as Alclad. The pure aluminum cladding acts as a sacrificial anode, providing cathodic protection to the copper- or zinc-rich core alloy. The thickness of the Alclad layer is typically 2.5% to 5% of the total sheet thickness per side. During structural repair operations, any machining, abrasive blending, or chemical milling that penetrates the Alclad layer exposes the highly reactive core alloy, necessitating the immediate application of a chemical conversion coating or anodic film to restore corrosion resistance. The heat treatment of aluminum alloys involves three critical phases: solution heat treatment, quenching, and aging. Solution heat treatment requires heating the alloy to a precise temperature (e.g., 900°F to 930°F for 2024) to dissolve the alloying elements into a solid solution. This is immediately followed by rapid quenching in cold water to retain a supersaturated solid solution at room temperature, resulting

in the highly formable but low-strength W temper. The final strength is achieved through precipitation hardening (aging), either naturally at room temperature (T3 or T4 temper) or artificially in a precision-controlled oven at elevated temperatures (e.g., 250°F for 7075-T6). The time-temperature profile during artificial aging must be strictly adhered to; under-aging results in sub-optimal yield strength, while over-aging degrades the tensile properties and alters the fracture toughness. Repair parts fabricated from annealed (O temper) sheet stock must be formed and subsequently heat-treated to the required T temper, a process that introduces significant distortion and requires specialized fixturing and subsequent dimensional inspection.

Chapter 3

Titanium and High-Strength Steel Alloy Fabrication and Repair Constraints

Titanium alloys, primarily Ti-6Al-4V (Grade 5) and Ti-3Al-2.5V, are utilized in high-temperature zones, highly corrosive environments, and critical structural joints where high specific strength and exceptional fatigue life are required. Ti-6Al-4V is an alpha-beta alloy, where the aluminum stabilizes the hexagonal close-packed (HCP) alpha phase, providing high-temperature creep resistance, and the vanadium stabilizes the body-centered cubic (BCC) beta phase, enabling heat treatment and improving ductility. The fabrication and repair of titanium structures present severe metallurgical and machining challenges. Titanium exhibits a strong chemical affinity for oxygen and nitrogen at elevated temperatures; therefore, any welding or thermal cutting operations must be performed in an inert argon atmosphere to prevent alpha-case formation, a brittle, oxygen-enriched surface layer that severely degrades fatigue life and must be completely removed via chemical milling or aggressive mechanical machining prior to service. Furthermore, titanium is highly susceptible to galling and cold-welding during machining and fastener installation due to its low thermal conductivity and high friction coefficient. Machining requires rigid setups, extremely sharp carbide tooling, low cutting speeds, high feed rates, and copious amounts of high-pressure coolant to prevent work hardening and tool failure. During fastener installation, titanium bolts must be heavily lubricated with dry film lubricants or specialized anti-seize compounds to prevent galling within the nut or threaded hole.

High-strength low-alloy (HSLA) steels, such as 300M, 4340M, and 4330M, are utilized in landing gear components, engine mounts, and critical structural fittings where ultimate tensile strengths exceeding 280,000 psi are required. The primary vulnerability of these ultra-high-strength steels is hydrogen embrittlement, a catastrophic failure mechanism where atomic hydrogen diffuses into the metal lattice, accumulating at regions of high triaxial stress and causing sudden, brittle fracture without macroscopic plastic deformation. Hydrogen can be introduced during electroplating, chemical stripping, corrosion removal, or acidic cleaning processes. To mitigate this, any high-strength steel component subjected to these processes must undergo a hydrogen embrittlement relief bake within four hours of the hydrogen-introducing operation. The standard baking protocol requires heating the component to $375^{\circ}\text{F} \pm 25^{\circ}\text{F}$ for a minimum of 24 hours to facilitate the out-diffusion of the trapped hydrogen. Additionally, the use of

cadmium plating for corrosion protection on high-strength steels is heavily restricted due to the risk of solid metal embrittlement if the cadmium melts or is subjected to high stress in the presence of micro-cracks. Alternative protective coatings, such as ion vapor deposited (IVD) aluminum or specialized zinc-nickel alloy plating, are increasingly mandated by structural repair manuals to provide sacrificial corrosion protection without the embrittlement risks associated with cadmium. The machining of HSLA steels requires strict control of grinding parameters to prevent grinding burn, a localized thermal damage mechanism that induces untempered martensite formation and severe tensile residual stresses, leading to immediate surface cracking. Nital etch inspection (ASTM E1417) is mandatory following any grinding operation on landing gear or high-strength steel fittings to detect these thermal damage zones.

Chapter 4

Composite Material Architecture, Cure Kinetics, and Damage Tolerance

Advanced composite structures, governed by ATA 51 and specific manufacturer composite repair manuals, utilize Carbon Fiber Reinforced Polymers (CFRP) and Glass Fiber Reinforced Polymers (GFRP) embedded in thermosetting epoxy or bismaleimide (BMI) matrices. The structural efficiency of CFRP is derived from the high axial stiffness and strength of the carbon filaments, which are highly anisotropic. To achieve quasi-isotropic in-plane mechanical properties, the plies are oriented in a symmetric and balanced layup sequence, typically $[0/\pm 45/90]_s$. The 0-degree plies carry axial bending and tensile loads, the ± 45 -degree plies resist shear loads and provide damage tolerance, and the 90-degree plies provide transverse stability and Poisson's ratio control. The consolidation and curing of these laminates require precise control of temperature, pressure, and vacuum to achieve the target fiber volume fraction (typically 60% to 65%) and to minimize void content. Autoclave curing applies an isostatic pressure of up to 100 psi to consolidate the plies and suppress the expansion of entrapped volatiles, while a vacuum of 22 to 29 inches of mercury extracts moisture and low-molecular-weight outgassing products. The cure kinetics of the epoxy matrix are highly exothermic; the heat generated by the cross-linking reaction must be carefully managed by the autoclave ramp rates to prevent thermal runaway, which can cause matrix degradation and severe internal residual stresses. The degree of cure α is monitored using embedded dielectric sensors or differential scanning calorimetry (DSC) of witness coupons, ensuring the glass transition temperature (T_g) meets the minimum specification, typically 250°F to 350°F for aerospace epoxies.

Sandwich panel construction, utilizing thin CFRP or aluminum face sheets bonded to a lightweight Nomex (aramid paper) or aluminum honeycomb core, provides exceptional bending stiffness with minimal mass penalty. The structural integrity of sandwich panels relies heavily on the flatwise tensile strength and the shear strength of the core-to-face sheet adhesive bond. Damage to sandwich structures frequently manifests as core crush, where the cellular core buckles under localized impact or compressive loads, or as face sheet-to-core disbond, which drastically reduces the local buckling resistance of the face sheet. Repairing composite structures requires the complete removal of the damaged material, typically utilizing high-speed pneumatic routers with carbide burrs or diamond-coated abrasive wheels, to create a precise repair cavity. For solid laminates, the repair boundary is prepared using a stepped-lap or scarf joint geometry. The scarf ratio, defined as the ratio of the horizontal taper length to the vertical ply thickness, is strictly

mandated by the SRM, typically ranging from 1:20 to 1:50 for primary flight structures. A 1:50 scarf ratio ensures that the shear stress within the adhesive bondline is minimized and the load transfer between the parent laminate and the repair patch is gradual, preventing premature peel failure. The repair plies must be cut to exact dimensions, indexed to the correct fiber orientation, and consolidated using localized heat blankets and vacuum bags, requiring precise thermocouple placement to ensure the repair zone reaches the required cure temperature without overheating the surrounding parent structure.

Chapter 5

Non-Destructive Testing (NDT) Protocols for Structural Integrity Verification

Non-Destructive Testing (NDT) is the fundamental mechanism for verifying the structural integrity of airframe components and validating the quality of executed repairs without inducing further material degradation. Eddy Current Testing (ET) is the primary modality for inspecting conductive materials, particularly aluminum alloys, for surface and near-surface fatigue cracks, as well as for measuring the thickness of non-conductive coatings and the conductivity of heat-treated alloys. The physical principle relies on inducing an alternating magnetic field into the test material via a probe coil, which generates circulating eddy currents. Any discontinuity, such as a crack or corrosion pitting, disrupts the flow of these currents, altering the impedance of the probe coil. The depth of penetration, or standard skin depth δ , is governed by the equation $\delta = \frac{1}{\sqrt{\pi f \mu \sigma}}$, where f is the excitation frequency, μ is the magnetic permeability, and σ is the electrical conductivity. To detect deep sub-surface flaws in thick sections, low-frequency eddy current probes are utilized, while high-frequency probes are required for high-resolution surface inspections. Calibration of the ET equipment must be performed using reference standards manufactured from the same alloy and heat treatment as the test article, containing precisely machined artificial flaws, such as electrical discharge machined (EDM) notches of known depth and length.

Ultrasonic Testing (UT) utilizes high-frequency acoustic waves (typically 1 MHz to 10 MHz) to detect internal flaws, measure material thickness, and evaluate the bond integrity of composite and metallic sandwich structures. Pulse-echo UT is standard for inspecting monolithic metallic structures, where the time-of-flight of the reflected acoustic wave indicates the depth of the flaw. For inspecting complex geometries or welds, angle-beam shear wave UT is employed, requiring the application of Snell's Law to calculate the precise refraction angle of the acoustic wave as it transitions from the plastic wedge into the metallic test piece. In composite structures, UT is critical for detecting delaminations, porosity, and disbonds. The attenuation of the acoustic signal is directly proportional to the void content within the composite matrix; a high void content scatters the ultrasonic energy, resulting in a significant reduction in the amplitude of the back-wall echo. Radiographic Testing (RT) utilizes X-rays or gamma rays to inspect internal structures, particularly complex castings, welds, and multi-layered assemblies where

UT is ineffective. The sensitivity of RT is verified using Image Quality Indicators (IQIs), or penetrameters, which must be placed on the source side of the test object. The geometric unsharpness U_g of the radiograph is calculated as $U_g = \frac{f \cdot t}{D}$, where f is the focal spot size of the X-ray tube, t is the distance from the source side of the object to the film, and D is the source-to-object distance. To ensure high-resolution imaging, D must be maximized and f minimized, strictly adhering to the exposure charts and density requirements specified in ASTM E1742. Fluorescent Penetrant Inspection (FPI) and Magnetic Particle Inspection (MPI) are reserved for surface-breaking flaws, requiring rigorous surface preparation, precise dwell times, and controlled ultraviolet illumination to ensure the detection of micro-cracks that are invisible to the naked eye.

Chapter 6

Fastener Selection, Hole Preparation, and Interference Fit Mechanics

The structural integrity of airframe joints relies on the precise selection, installation, and load transfer mechanics of aerospace fasteners, which are manufactured to stringent Military Standard (MS), National Aerospace Standard (NAS), and Boeing/NASA (BAC/BACN) specifications. Fastener materials are selected based on the parent material to prevent galvanic corrosion and to match the thermal expansion coefficients. Titanium fasteners (e.g., Ti-6Al-4V) are standard for composite and titanium structures, while A286 iron-based superalloy or Inconel 718 fasteners are mandated for high-temperature zones and high-strength steel joints. Aluminum fasteners are strictly limited to secondary, lightly loaded interior structures due to their low shear strength and high susceptibility to stress corrosion cracking. The preparation of the fastener hole is a critical process that directly dictates the fatigue life of the joint. For close-tolerance bolts, the hole diameter tolerance is strictly maintained at $+0.0005 / -0.0000$ inches. The hole surface finish must not exceed 125 micro-inches Ra, and any burrs, smears, or delaminations at the hole exit must be meticulously removed using micro-shaving tools or precision reamers. In composite structures, hole drilling requires specialized diamond-coated or solid carbide "dagger" drill bits, strict feed-rate control, and backup blocks to prevent exit-side delamination and fiber pull-out.

Interference fit fasteners are extensively utilized in primary metallic structures to significantly enhance the fatigue life of the joint. The installation of an interference fit bolt or sleeve induces a radial compressive residual stress field around the fastener hole, which counteracts the applied cyclic tensile loads and retards fatigue crack initiation. The radial interference I is typically specified as 1% to 3% of the hole diameter. The resulting radial stress σ_r at the edge of the hole can be approximated using the thick-walled cylinder equations derived from Lamé's theory, demonstrating that the compressive stress decays exponentially with distance from the hole. However, excessive interference can cause localized yielding, hole elongation, or, in composite materials, severe internal delamination and matrix crushing. To achieve high interference levels without damaging the hole surface, specialized installation techniques such as force-controlled hydraulic presses or ultrasonic vibration-assisted insertion are employed. Countersinking is required for flush fastener installations to maintain aerodynamic smoothness. The depth of the countersink is strictly limited to prevent "knife-edge" conditions, where the remaining material thickness beneath the countersink head is insufficient to bear the clamp-up loads, leading to pull-through failure. The minimum remaining thickness (keel) is typically mandated to be at

least 0.005 inches for thin sheets, or a specific percentage of the sheet thickness as defined in the SRM. For thin sheets where a standard countersink would penetrate the material, dimpling (using a male/female die set) or micro-shaving techniques are utilized to create a flush profile without compromising the structural cross-section.

Chapter 7

Riveting Techniques, Flush Countersinking, and Solid/Shank Fastener Installation

Solid rivets, such as the AN470 (universal head) and AN426/MS20426 (100-degree flush head) series, remain the primary fastening method for metallic airframe skin splices and internal structural connections. The installation of solid rivets requires the precise application of force to upset the driven end, forming a shop head that mechanically locks the joint and induces a beneficial clamp-up force. The dimensional requirements for the driven shop head are strictly defined: the width must be 1.5 times the nominal shank diameter ($1.5D$), and the height must be 0.5 times the nominal shank diameter ($0.5D$). Deviations from these dimensions, such as undersized, cracked, or "smiley" shop heads, result in inadequate clamp-up and reduced shear strength, mandating immediate drill-out and replacement. Riveting is performed using either a pneumatic rivet hammer and a bucking bar, or a pneumatic/hydraulic rivet squeezer. Squeezing is the preferred method as it provides uniform, controlled deformation, minimizes acoustic vibration, and reduces the risk of work-hardening the rivet shank. When using a rivet hammer, the technician must precisely match the hammer intensity and the bucking bar mass to the rivet diameter and material to prevent over-driving or damaging the parent sheet.

High-strength structural fasteners, such as Hi-Loks (internally threaded pins with a collar) and lockbolts (swaged collars over a grooved pin), are utilized in primary load-bearing joints where high clamp-up and precise preload are required. The installation of Hi-Loks requires the application of a specific installation torque, followed by the tightening of the collar until the hexagonal wrench flats shear off at a pre-calibrated torque value, ensuring a consistent and verifiable clamp-up load. The hole preparation for Hi-Loks is critical; they are designed for light interference fits, and the holes must be reamed to exact tolerances. Blind fasteners, such as CherryMAX or Huck rivets, are mandated for closed structures where access to the back side of the joint is impossible. These fasteners utilize a mechanical locking mechanism and a built-in breaking stem to form a blind head. The installation requires specialized pulling tools that must be held perfectly perpendicular to the work surface to prevent asymmetric blind head formation and stem retention failures. The inspection of installed fasteners involves visual verification of head seating, shop head dimensions, and the presence of mechanical locking features. Any fastener that is proud of the flush surface by more than 0.003 inches, or recessed by more than

the allowable limits specified in the SRM, must be removed. The removal of defective fasteners requires extreme precision; the drill bit must be perfectly centered on the fastener head, and the drilling must be stopped exactly at the shear plane to prevent enlarging the parent hole or scoring the faying surfaces, which would necessitate an oversize repair.

Chapter 8

Structural Bonding: Surface Preparation, Adhesive Chemistry, and Cure Profiles

Structural adhesive bonding is a critical joining methodology that distributes loads continuously across the faying surface, eliminating the stress concentrations inherent in mechanical fastening and providing superior fatigue performance and aerodynamic smoothness. The structural integrity of a bonded joint is fundamentally dictated by the surface preparation of the adherends, as adhesive failure (disbond at the interface) is vastly more common and catastrophic than cohesive failure (failure within the adhesive matrix). For aluminum structures, the standard surface preparation involves alkaline cleaning, followed by phosphoric acid anodizing (PAA) or the application of a chemical conversion coating such as Pasa-Jell 105. PAA creates a highly porous, micro-columnar aluminum oxide layer that provides exceptional mechanical interlocking and hydration resistance for the adhesive. The efficacy of the surface preparation is verified using the water break test, where a film of deionized water must remain continuous and unbroken on the surface for a minimum of 30 seconds, indicating a high surface energy and the complete removal of hydrophobic contaminants. For composite adherends, surface preparation typically involves the removal of the mold release agent via abrasive grit blasting (using aluminum oxide or plastic media) followed by a rigorous solvent wipe using a two-cloth method with MEK or acetone, strictly adhering to environmental and safety regulations.

The adhesives utilized in primary aerospace structures are predominantly two-part, high-temperature curing epoxies or film adhesives (e.g., 3M Scotch-Weld or Cytec FM series). Film adhesives are highly preferred for large-area bonding due to their precise thickness control, consistent resin-to-curing-agent ratio, and ease of handling. The cure profile of the adhesive is strictly governed by the manufacturer's technical data sheet, requiring a specific ramp rate, hold temperature (typically 250°F or 350°F), and dwell time under vacuum or autoclave pressure. The bondline thickness is a critical design parameter; it must be thick enough to accommodate the shear strain differential between the adherends, but thin enough to prevent excessive peel stresses and void formation. To control the bondline thickness, the film adhesive is often manufactured with embedded glass beads or a woven nylon carrier scrim, typically maintaining a bondline of 0.003 to 0.008 inches. The mechanical performance of the bonded joint is quantified through lap shear testing (ASTM D1002) and peel testing (ASTM D1876). Furthermore, the environmental

durability of the bond is verified using the wedge crack test (ASTM D3762), where a bonded specimen is forced open with a wedge and exposed to high humidity; the rate of crack propagation along the bondline must remain below strict thresholds to certify the adhesive system for primary structure. During repair operations, the application of the adhesive, the assembly of the joint, and the cure cycle must be meticulously documented, and the repair must be inspected using ultrasonic testing or coin-tap testing to verify the absence of disbonds, porosity, or inadequate fillet formation at the edges.

Chapter 9

Sheet Metal Forming, Bending Allowances, and Jogging Calculations

The fabrication of sheet metal repair parts requires a rigorous understanding of material deformation mechanics, bend allowances, and springback to ensure the final part conforms precisely to the aerodynamic and structural contours of the parent airframe. When a flat sheet is bent, the material on the outside of the bend radius is subjected to tensile stress, while the material on the inside is subjected to compressive stress. The neutral axis, the theoretical plane within the sheet thickness that experiences zero strain, shifts inward toward the inside of the bend as the bend radius decreases. The exact location of the neutral axis is defined by the K-factor, which is the ratio of the distance from the inside bend surface to the neutral axis, divided by the material thickness T . The Bend Allowance (BA), which represents the arc length of the neutral axis through the bend, is calculated using the formula $BA = \frac{\pi}{180} \times \theta \times (R + K \times T)$, where θ is the bend angle in degrees and R is the inside bend radius. Accurate calculation of the BA is essential for determining the correct flat pattern layout; an error in the BA will result in a formed part that is either too long or too short, compromising the fastener hole alignment and the structural fit.

The minimum allowable bend radius for a specific material and thickness is strictly dictated by the material's ductility and the heat treatment temper. Bending a sheet tighter than the minimum bend radius will induce severe tensile cracking on the outside of the bend or compressive wrinkling on the inside. For example, 2024-T3 aluminum must be bent in the annealed (O) or W temper and subsequently heat-treated, or it must be bent transversely to the grain direction with a significantly larger bend radius compared to bending parallel to the grain. Following the bending operation, the material exhibits springback, an elastic recovery that causes the bend angle to open slightly upon removal of the forming load. To compensate for springback, the sheet must be over-bent by a calculated angle, which is determined empirically based on the material's yield strength, thickness, and the bend radius. Jogging, the process of forming an offset in the sheet to maintain a flush surface at a lap joint, introduces complex localized stresses. The joggle depth is typically limited to a maximum of the sheet thickness, and the joggle length must be sufficient to prevent severe localized thinning and tearing. The neutral axis shifts dramatically within the joggle transition zone, requiring the addition of relief cuts or radiused corners to prevent crack initiation. The forming of complex contours, such as those required for fuselage skin repairs or wing leading edges, necessitates the use of precision-machined male and female die sets, drop

hammers, or hydro-presses, combined with intermediate annealing cycles for hardening alloys to prevent work-hardening and subsequent fracture during the forming process.

Chapter 10

Corrosion Identification, Chemical Treatment, and Preventative Coatings

Corrosion is a pervasive degradation mechanism that compromises the structural integrity, reduces the load-bearing cross-section, and acts as a severe stress concentrator that accelerates fatigue crack initiation. Aerospace structures are susceptible to multiple corrosion morphologies, including surface pitting, intergranular corrosion, exfoliation, fretting, and stress corrosion cracking (SCC). Intergranular corrosion attacks the grain boundaries of heat-treated aluminum alloys, while exfoliation corrosion propagates parallel to the surface along the elongated grain structure of extruded or rolled products, causing the metal to delaminate and swell with immense force, potentially deforming the surrounding structure. Fretting corrosion occurs at the faying surfaces of mechanically fastened joints subjected to microscopic relative motion, generating hard, abrasive aluminum oxide debris that accelerates wear and initiates fatigue cracks. The identification and quantification of corrosion damage are critical; the depth of material loss must be measured using precision depth gauges or ultrasonic thickness testers and compared against the Allowable Damage Limits (ADL) defined in the SRM. If the corrosion exceeds the ADL, the material must be blended out within strict slope limits, or the component must be repaired or replaced.

The mitigation and prevention of corrosion require the meticulous application of chemical conversion coatings and protective primers. Chemical conversion coatings, such as Alodine 1201 (hexavalent chromium) or modern trivalent chromium and zirconium-based alternatives (e.g., Iridite), are applied to bare aluminum surfaces to create a micro-crystalline, electrically conductive oxide layer that passivates the surface and provides an excellent anchor profile for subsequent primer adhesion. The application process requires strict control of the chemical concentration, pH, temperature, and dwell time, followed by a thorough deionized water rinse to halt the chemical reaction. For components requiring enhanced corrosion resistance and surface hardness, anodizing is employed. Sulfuric acid anodizing (SAA) and boric-sulfuric acid anodizing (BSAA) create a thick, porous aluminum oxide layer that is subsequently sealed in boiling water or a nickel acetate solution to close the pores and lock in corrosion-inhibiting dyes. Following surface treatment, the components are coated with high-performance primers, typically two-part epoxy-polyamide systems. Legacy primers utilized strontium chromate pigments for their exceptional active corrosion inhibition; however, due to severe environmental and health regulations, modern non-chromate primers utilizing ion-exchange pigments or rare-earth

inhibitors are now mandated. The final topcoats, typically polyurethane-based, provide UV resistance, chemical resistance to hydraulic fluids and de-icing agents, and the required aerodynamic gloss. Furthermore, the faying surfaces of all structural joints must be sealed using polysulfide or polyurethane sealants (e.g., PRC-DeSoto) to prevent the ingress of moisture and electrolytes, thereby eliminating the environment required for galvanic and crevice corrosion.

Chapter 11

Damage Assessment, Allowable Damage Limits (ADL), and Repair Design Margins

The assessment of structural damage and the subsequent design of the repair require a rigorous analytical approach to ensure the restored structure meets the ultimate load, limit load, and damage tolerance requirements. The Structural Repair Manual (SRM) provides the Allowable Damage Limits (ADL) for various airframe components, defining the maximum size of dents, scratches, gouges, and punctures that can be accepted without repair, or the maximum material loss that can be restored via simple blending. For composite structures, the assessment of Barely Visible Impact Damage (BVID) is particularly critical. BVID is defined as damage that leaves an indentation of less than 0.100 inches (often difficult to detect during a routine visual walk-around) but may cause extensive internal delamination and matrix cracking, severely degrading the Compression After Impact (CAI) strength. The SRM mandates the use of ultrasonic C-scan or tap testing to map the true extent of the internal delamination, which frequently extends far beyond the visible surface impact zone. The repair design must completely encompass this internal damage zone with a sufficient margin.

When designing a bonded or bolted doubler repair, the engineer must calculate the load transfer mechanics, accounting for the shear lag phenomenon. In a bonded lap joint, the shear stress is not distributed uniformly across the overlap length; it peaks heavily at the edges of the overlap and drops to near zero in the center. This non-uniform stress distribution is governed by the shear lag parameter $\lambda = \sqrt{\frac{G_a}{t_a} \left(\frac{1}{E_1 t_1} + \frac{1}{E_2 t_2} \right)}$, where G_a and t_a are the shear modulus and thickness of the adhesive, and E_1, t_1, E_2, t_2 are the modulus and thickness of the adherends. To prevent premature peel failure at the edges of the doubler, the repair patch must be tapered or "feathered" at the extremities to gradually reduce the stiffness mismatch and minimize the peel stresses. For mechanically fastened repairs, the joint must be analyzed for multiple failure modes: net-tension failure across the fastener holes, shear-out failure, bearing failure of the hole, and fastener shear failure. The bearing bypass stress, which represents the complex interaction between the load carried by the fastener (bearing) and the load carried by the net section of the sheet (bypass), must be plotted on a bearing-bypass interaction diagram to verify that the combined stresses remain within the safe envelope. The repair design must also account for the eccentricity introduced by single-sided doubler patches, which induces secondary bending

moments under tensile loading, requiring the use of double-sided (symmetric) patches whenever aerodynamic and structural access permits.

Chapter 12

Airworthiness Documentation, Logbook Entries, and Structural Repair Manual (SRM) Compliance

The legal airworthiness of an aircraft following a structural repair is inextricably linked to the meticulous execution of regulatory documentation, as mandated by 14 CFR Part 43. A structural repair is classified as a major repair if it might appreciably affect weight, balance, structural strength, performance, powerplant operation, flight characteristics, or other qualities affecting airworthiness. Major repairs must be performed in accordance with FAA-approved data, which includes the aircraft manufacturer's Structural Repair Manual (SRM), Supplemental Type Certificates (STC), or specific engineering data approved by a Designated Engineering Representative (DER) via FAA Form 8110-3. Upon completion of the repair, the certifying mechanic or inspector must execute a detailed maintenance logbook entry in accordance with 14 CFR 43.9. This entry must explicitly describe the exact nature of the damage, the specific SRM chapter, figure, and revision level utilized for the repair design, the materials and fasteners installed (including part numbers and material specifications), and the results of all mandatory Non-Destructive Testing (NDT) and functional checks. The entry must be signed, dated, and include the certificate number of the authorized individual returning the aircraft to service.

Furthermore, major repairs require the completion of FAA Form 337 (Major Repair and Alteration), which serves as the permanent, centralized record of the structural modification. Block 8 of the Form 337 must contain a comprehensive description of the work performed, and Block 7 must detail the specific approved data utilized. If the repair deviates from the pre-approved SRM limits, or if the damage exceeds the scope of the SRM, the operator must submit a structural repair request to the aircraft manufacturer's engineering department or an authorized DER. The resulting custom repair design will include specific instructions for material preparation, hole patterns, fastener types, torque values, and post-repair inspection intervals. The traceability of all structural materials and fasteners is paramount; every piece of raw material, pre-preg composite, and structural fastener must be accompanied by a Certificate of Conformance (C of C) or material test report that verifies the chemical composition, heat treatment, and mechanical properties against the aerospace material specifications (AMS). The heat lot numbers and cure batch numbers must be recorded in the aircraft's permanent structural modification records to ensure that, in the event of a fleet-wide material defect or airworthiness

directive, the specific components can be accurately identified and tracked. Failure to maintain this rigorous chain of documentation renders the structural repair legally invalid, compromises the damage tolerance tracking of the airframe, and constitutes a severe violation of federal aviation regulations, potentially resulting in the immediate grounding of the aircraft.

Appendix D: Daily Airframe Structural Repair Checklist

This checklist governs sheet metal forming, rivet patterns, hole tolerances, and sealant processing limits during structural alterations and doubler patch repairs.

Table 12.1: Airframe Metallic Structures and Fastener Specifications

Maintenance Item	Measurement / Range	Regulatory / Tool Ref
Rivet Edge Distance (Minimum)	$2.0 \times D_{\text{rivet}}$	Distance from center of hole
Rivet Edge Distance (Maximum)	$4.0 \times D_{\text{rivet}}$	Distance from center of hole
Rivet Pitch (Spacing Between Rivets)	$3.0 \times D_{\text{rivet}}$ to $4.0 \times D_{\text{rivet}}$	Prevent sheet buckling or tearing
Formed Shop Head Diameter (D_{shop})	$1.5 \times D_{\text{nominal}}$	Dial caliper check
Formed Shop Head Height (H_{shop})	$0.5 \times D_{\text{nominal}}$	Dial caliper check
Stop-Drill Crack Tip Diameter	0.125 inches (1/8-inch reamer)	#30 drill bit centered on crack
Sheet Metal Bend Setback	$SB = K \cdot (R + T)$	Setback calculation based on material
90-Degree Setback K-Factor	1.000	Standard bracket right-angle
45-Degree Setback K-Factor	0.414	Accessory bracket support l
Hi-Lok 3/16-inch Reamed Hole Diameter	0.1890 to 0.1900 inches	Close-tolerance interference
Hi-Lok Grip Length Tolerance	0.5 to 2.0 threads showing	Thread engagement beyond
Alodine 1201 Chemical Dwell Time	2.0 to 5.0 minutes	Acid prep for aluminum pai
Zinc Chromate Drying Time	1.0 to 2.0 hours	TT-P-1757 safety primer ba
Class B-1/2 Sealant Work Life	≤ 30 minutes	Structural faying surface ap
Class B-1/2 Sealant Full Cure Time	24 hours	Curing before cabin pressur
Doubler Patch Material Thickness	1.0 gauge thicker than original skin	FAA Form 337 Major Repa